

Class 27 Math Induction

Goals

1. Learn the difference between weak and strong induction.
2. Practice several examples.

Example 1: Prove that $7^n - 1$ is a multiple of 6 for all $n \geq 1$

Conclusion: Since the statement is true for $n = 1$ and is true for $n = k+1$ whenever true for $n = k \geq 1$ we know the statement is true for all integers $n \geq 1$

Strong Induction: If $[P(a)$ is true for some starting integer a and if all the statements from $P(a)$ through $P(k)$ being true implies that $P(k+1)$ is true], then $P(n)$ is true for all $n \geq a$.

Example 2: Use strong induction to prove every integer greater than 1 is a prime or a product of primes. (existence part of the Fundamental Theorem of Arithmetic).

$P(n)$ is the statement " n is prime or a product of primes"

Conclusion: Since the statement is true for $n = 2$ and is true for $n = k+1$ whenever true for $n = 2$ through k , we know the statement is true for all integers $n \geq 2$

Example 3: The Fibonacci sequence is defined recursively by $f_1 = 1$, $f_2 = 1$, and $f_{n+2} = f_n + f_{n+1}$ for $n \geq 1$. Write out the first 8 terms of the sequence.

Use strong induction to prove $f_n < 2^n$ for all $n \geq 1$.

Conclusion: Since the statement is true for $n = 1$ and $n = 2$ and is true for $n = k+1$ whenever true for $n = 1$ through k , we know by strong induction the statement is true for all integers $n \geq 1$.

Example 4: Prove for all $n \geq 1$
$$\prod_{i=1}^n \frac{1}{(2i+1)(2i+2)} = \frac{2}{(2n+2)!}$$

Conclusion: Since the statement is true for $n = 1$ and is true for $n = k+1$ whenever true for $n = k$, we know by math induction the statement is true for all integers $n \geq 1$.

